Appendix B Kubernetes in production with DigitalOcean

This appendix covers

- Running a Kubernetes cluster on DigitalOcean
- Running a PostgreSQL database on DigitalOcean
- Running Redis on DigitalOcean
- Running RabbitMQ using a Kubernetes Operator
- Running Keycloak using a Helm chart

Kubernetes is the de facto standard for deploying and managing containerized workloads. We've been relying on a local Kubernetes cluster to deploy applications and services in the Polar Bookshop system throughout the book. For production, we need something else.

All major cloud providers offer a managed Kubernetes service. In this appendix, you'll see how to use DigitalOcean to spin up a Kubernetes cluster. We'll also rely on other managed services provided by the platform, including PostgreSQL and Redis. Finally, this appendix will guide you through the deployment of RabbitMQ and Keycloak directly in Kubernetes.

Before moving on, you need to ensure that you have a DigitalOcean account. When you sign up, DigitalOcean offers a 60-day free trial with a $100 credit that is more than enough to go through the examples in chapter 15. Follow the instructions on the official website to create an account and start a free trial (https://try.digitalocean.com/freetrialoffer).